Information Theory & Error Correcting Codes

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Disclaimer

This document contains the lectures given by Dr. Seffah.

> To separate the contents of the course to actual additions or out of context information, a black band will be added by its side like the one on this comment.

Contents

Chapter 0: Remainders 2
1. Congruences & $\mathbb{Z}/m\mathbb{Z}$ Arithmetic 2
2. Euler φ Function 2
3. Quadratic Residues 3
Chapter 1: Introduction To Information Theory 6
1.1. Entropy 7
1.1.1. Source Coding 7
1.2. Shannon Theorems 8
1.2.1. Noise And Errors 9
Chapter 2: Information & Entropy 10
2.1. The Communication System Model 10
2.1.1. Source & Message 10
2.1.2. Transmitter Processing 10
2.1.3. Receiver Processing 10
2.2. Quantifying Information 10
2.2.1. Fundamental Postulates Of Information 10
2.3. Entropy 10
2.3.1. Definition Of Entropy 10
2.3.2. Properties Of Entropy 10
2.3.3. Joint Entropy 11
2.3.4. Properties Of Joint Entropy 11

Chapter 0

Remainders

1. Congruences & $\mathbb{Z}/m\mathbb{Z}$ Arithmetic

Definition 1.1 (Congruence Of Integers / Congruence Class): *Let $a, b, m \in \mathbb{Z}$ with $m > 0$, we say that a is congruent to b modulo m and we write $a \equiv b \bmod m$ if $m \mid a - b$ which gives an equivalence relation. The class of a in the congruence relation by m is called the congruence class of a modulo m, which is $\overline{a} = a + m\mathbb{Z}$.*

Theorem 1.2: *Let $a, b, c, d \in \mathbb{Z}$ and $m \in \mathbb{N}$, with $a \equiv b \bmod m$ and $c \equiv d \bmod m$ then we have the following statements are true:*

1. $a + c \equiv b + d \bmod m$.
2. $a - c \equiv b - d \bmod m$.
3. $ac \equiv bd \bmod m$.

We denote $\mathbb{Z}/m\mathbb{Z}$, the set of all congruence classes modulo m.

2. Euler φ Function

Definition 2.3 (φ Function): *The Euler φ function is defined as $\varphi : \mathbb{Z} \to \mathbb{N}$, where $\varphi(n) = \#\{x \in [\![0, m-1]\!] \mid \gcd(x, m) = 1\}$.*

Lemma: *Let $r, m, p \in \mathbb{N}$ where p is prime, then $\gcd(m, p^r) \neq 1 \Leftrightarrow p \mid m$*

Proof. $\Rightarrow$ let $d = \gcd(m, p^r)$, since $d \mid p^r$ and p is prime then $d = p^k$ for some $k \in \mathbb{N}$, and $p^k = d \mid m$ thus $p \mid p^k$ and $p^k \mid m$ so $p \mid m$. $\Leftarrow$ suppose that $p \mid m$ and since $p \mid p^r$ too then $p \mid \gcd(m, p^r)$ given $p > 1$ then $\gcd(m, p^r) \neq 1$. $\square$

Proposition 2.4: *Let p be a prime number, then $\varphi(p) = p - 1$ and $\varphi(p^r) = p^r - p^{r-1} = p^r\left(1 - \frac{1}{p}\right)$.*

Proof. Let $p, r \in \mathbb{N}$ with p prime. From the lemma, we have that for $m \in \mathbb{N}$, $\gcd(p^r, m) = 1 \Leftrightarrow p \nmid m$, the numbers that are divisible by p in $[\![0, p^r - 1]\!]$ are of the form kp where $k \in [\![0, p^{r-1} - 1]\!]$, so the number of numbers that are divisible by p in $[\![0, p^r - 1]\!]$ is $p^r - p^{r-1}$ so $\varphi(p^r) = p^r - p^{r-1}$. $\square$

Theorem 2.5 (Lagrange): *Let G be a finite group and H a subgroup of G, then $\# H \mid \# G$.*

This theorem is already done multiple times in algebra so no need to prove it here.

Lemma 2.6: *Let $n \in \mathbb{N}$, $x \in (\mathbb{Z}/n\mathbb{Z})^*$ (invertible) $\Leftrightarrow \gcd(x, n) = 1$.*

Proof. $\Rightarrow$ Suppose that $x \in (\mathbb{Z}/n\mathbb{Z})^*$, then $\exists y \in \mathbb{Z}/n\mathbb{Z}$ such that $xy = 1 \bmod n$, then $xy = 1 + kn \Rightarrow xy - kn = 1$ by Bezout theorem we have $\gcd(x, n) = 1$. $\Leftarrow$ SUppose that $\gcd(x, n) = 1$, by Bezout theorem, $\exists y, k \in \mathbb{Z}, xy + kn = 1$ and thus $xy + kn = 1 \bmod n \Rightarrow xy = 1 \bmod n$ so $x \in (\mathbb{Z}/n\mathbb{Z})^*$. $\square$

We need this lemma since $\varphi(n) = \#(\mathbb{Z}/n\mathbb{Z})^*$.

Theorem 2.7 (Euler): *Let m be a positive integer modulo a be an integer relatively prime to m then $a^{\varphi(m)} \equiv 1 \bmod m$.*

Proof. Let a relatively prime to m, then $\gcd(a, m) = 1 \Rightarrow a \in (\mathbb{Z}/m\mathbb{Z})^*$, a has finite order thus $\exists k \in \mathbb{N}, a^k = 1 \bmod m$, $\langle a \rangle$ is a subgroup of $(\mathbb{Z}/m\mathbb{Z})^*$ so $k \mid \#(\mathbb{Z}/m\mathbb{Z})^* = \varphi(m)$ and thus $\varphi(m) = kb$, thus $a^{\varphi(m)} = a^{kb} = \left(a^k\right)^b = 1^b = 1 \bmod m$, which completes the proof. $\square$

Corollary 2.8 (Fermat): *Let p be a prime, if the integer a is not divisible by p then $a^p \equiv a \bmod p$.*

Proof. Just take $m = p$ in Euler's theorem, then $\varphi(m) = p - 1$ so $a^{\varphi(m)} = a^{p-1} = 1 \bmod m$, so $a^p = a \bmod m$. □

3. Quadratic Residues

Definition 3.9 (Quadratic Residue): *Let p be an odd prime and a an integer not divisible by p, we say that a is a quadratic residue modulo p if there exists $x \in \mathbb{Z}$, such that $x^2 \equiv a \bmod p$.*

Theorem 3.10: *An integer a is a quadratic residue modulo p if and only if $\gcd(a, p) = 1$ and a has a square rest modulo p.*

Proof. $\Rightarrow$ Suppose that a is a quadratic residue modulo p, that is, there exists $x \in \mathbb{Z}$ such that $x^2 \equiv a \bmod p$. If $\gcd(a, p) \neq 1$ then $p \mid a$ so $a \equiv 0 \bmod p$ thus $x = 0$, so necessarily $\gcd(a, p) = 1$ and a has x^2 remainder when divided by p thus the rest is a perfect square. $\Leftarrow$ Suppose now that $\gcd(a, p) = 1$ and a has a square rest modulo p, then $a \not\equiv 0 \bmod p$ and $a \equiv x^2 \bmod p$ thus a is quadratic residue. □

Definition 3.11 (Legendre Symbol): *Let p be an odd prime, define the Legendre symbol as:*

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } \gcd(a,p) = 1 \text{ and } a \text{ is a quadratic residue} \bmod p \\ -1 & \text{if } \gcd(a,p) = 1 \text{ and } a \text{ is not a quadratic residue} \\ 0 & \text{if } p \text{ divides } a \end{cases}$$

Theorem 3.12: *Let p be an odd prime, for every integer a*

$$\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \bmod p$$

Proof. We treat by cases

1. $p \mid a$: then $a = 0 \bmod p \Rightarrow a^{\frac{p-1}{2}} = 0^{\frac{p-1}{2}} = 0 \bmod p$.
2. $p \nmid a$: then $\gcd(a, p) = 1$, by Fermat's little theorem we have that $a^{p-1} \equiv 1 \bmod p$, the square roots of 1 are -1 and 1 only thus $a^{\frac{p-1}{2}} \equiv 1$ or $-1 \bmod p$.
   1. If a is a quadratic residue then $\exists x \in \mathbb{Z}, x^2 \equiv a \bmod p \Rightarrow a^{\frac{p-1}{2}} \equiv x^{2\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \bmod p$.
   2. If $a^{\frac{p-1}{2}} \equiv 1 \bmod p$, we have that $\mathbb{Z}_p$ is a finite field thus there exists a generator g of $(\mathbb{Z}_p)^*$ (primitive element theorem in finite fields), then $a \equiv g^k \bmod p$, we have then $a^{\frac{p-1}{2}} \equiv g^{k\frac{p-1}{2}} \equiv 1 \bmod p$. Given that the generator g has order $p - 1$ then $k\frac{p-1}{2} \equiv 0 \bmod p - 1$ thus $k\frac{p-1}{2} = m(p-1) \Rightarrow k = 2m$ so k is even, thus $a \equiv g^{2m} = (g^m)^2 \bmod p$ so a is a quadratic residue modulo p.

Given that a is a quadratic residue modulo p if and only if $a^{\frac{p-1}{2}} \equiv 1 \bmod p$ and $a^{\frac{p-1}{2}} \equiv 1$ or $-1 \bmod p$ then necessarily if a is not a quadratic residue, $a \equiv -1 \bmod p$. □

Exercise 3.13:

1. *Decompose into partial fractions in $\mathbb{R}[x]$ the rational function*
$$\frac{x}{x^4 + x^2 + 1}$$
2. *Let K be a commutative field, and let $p : x^2 + \lambda x + \mu$ be a monic polynomial of degree 2, show that p is reducible over K if and only if it has a root in K.*
3. *Let $K = \mathbb{Z}/5\mathbb{Z}$ be the field of residue classes, factor the polynomial $(x^2 + 4)(x^2 + 3)$ into irreducible factors over K.*
4. *Still with $K = \mathbb{Z}/5\mathbb{Z}$, decompose into partial fractions the rational function*
$$\frac{x - \overline{2}}{(x^2 + \overline{4})(x^2 + \overline{3})}$$

1. Decomposing into partial fractions: we have that the denominator decomposes into the following two irreducible polynomials

$$\begin{aligned} x^4 + x^2 + 1 &= x^4 + 2x^2 + 1 - x^2 \\ &= (x^2+1)^2 - x^2 \\ &= (x^2+x+1)(x^2-x+1) \end{aligned}$$

and thus we have that the fraction decomposes as follows

$$\frac{x}{x^4+x^2+1} = \frac{ax+b}{x^2+x+1} + \frac{cx+d}{x^2-x+1}$$

by multiplying both sides by x^4+x^2+1 we get

$$\begin{aligned} &x = (ax+b)(x^2-x+1) + (cx+d)(x^2+x+1) \\ &0 = ax^3 - ax^2 + ax + bx^2 - bx + b + cx^3 + cx^2 + cx + dx^2 + dx + d - x \\ &0 = (a+c)x^3 + (-a+b+c+d)x^2 + (a-b+c+d-1)x + (b+d) \end{aligned}$$

$$\Rightarrow \begin{cases} a+c &= 0 \\ -a+b+c+d &= 0 \\ a-b+c+d-1 &= 0 \\ b+d &= 0 \end{cases} \Rightarrow \begin{cases} a = 0 \\ b = -1/2 \\ c = 0 \\ d = 1/2 \end{cases}$$

thus we get

$$\frac{x}{x^4+x^2+1} = \frac{1}{2(x^2-x+1)} - \frac{1}{2(x^2+x+1)}$$

2. Let K be a commutative field and let $P(x) = x^2 + \lambda x + \mu$. $\Rightarrow$ Suppose that $P(x)$ is reducible, then $P(x) = Q(x) \cdot R(x)$ with $\deg(Q) \neq 0$ and $\deg(R) \neq 0$ and since $\deg(Q) + \deg(R) = \deg(P)$ then necessarily $\deg(Q) = \deg(R) = 1$, so $Q(x) = ax + b$ and $R(x) = cx + d$ and since $a, b \in K$ then $-\frac{a}{b} \in K$ which is a root of Q and thus a root of P so P has a root in K. $\Leftarrow$ Suppose that $P(x)$ has a root α, then $(x - \alpha)$ divides P and thus we get that P is reducible.
3. Let $K = \mathbb{Z}/5\mathbb{Z}$, factor $(x^2+4)(x^2+3)$ into irreducible factors over K, $-3 = 2 \bmod 5$ is not a quadratic residue since $2^{\frac{5-1}{2}} = 4 = -1 \bmod 5$ thus x^2+3 is irreducible given it has no roots. While $-4 = 1$ is a quadratic residue with roots -1 and 1 so $(x^2+4)(x^2+3) = (x^2-1)(x^2+3) = (x-1)(x+1)(x^2+3)$.
4. Decomposing now the partial fraction in $\mathbb{Z}/5\mathbb{Z}$,

$$\begin{aligned} \frac{x-2}{(x^2+4)(x^2+3)} &= \frac{x-2}{(x+4)(x+1)(x^2+3)} \\ &= \frac{a}{x+4} + \frac{b}{x+1} + \frac{cx+d}{x^2+3} \end{aligned}$$

by multiplying all parts by $(x^2+4)(x^2+3)$ we get

$$x - 2 = a(x+1)(x^2+3) + b(x+4)(x^2+3) + (cx+d)(x+4)(x+1)$$

we set the following values for x

- $x = -1$: $-3 = b(-1+4)(1+3) = 12b = 2b \Rightarrow b = 3 * (-3) = -9 = 1$.
- $x = +1$: $-1 = a(1+1)(1+3) = 8a = 3a \Rightarrow a = 2 * (-1) = -2 = 3$.

Exercise 3.14:

1. *Decompose* 561 *into prime factors.*
2. *Let* a *be an indeterminate, for any integer* $n \geq 1$*, justify the formula*

$$a^n - 1 = (a-1)(a^{n-1} + a^{n-2} + \cdots + a + 1)$$

3. *Justify for any integer* $m \geq 1$ *and any integer* $n \geq 1$ *the formula*

$$a^{mn} - 1 = (a^m - 1)(a^{(n-1)m} + a^{(n-2)m} + \cdots + a^m + 1)$$

4. *Let* $a \in \mathbb{Z}$ *be an integer, prove* $a^{561} - a = a(a^{2\cdot 280} - 1) = a(a^2-1)m$ *for some integer* m.
5. *For any* $a \in \mathbb{Z}$*, show that* $a(a-1)(a+1)$ *is divisible by* 3.
6. *Show that* $a^{561} - a$ *is a multiple of* 3.
7. *Show that* $a^{561} - a$ *is a multiple of* 17.
8. *Show that* $a^{561} - a$ *is a multiple of* 11.
9. *Show that* $a^{561} \equiv a \bmod 561$.

1. The prime decomposition of 561 is $3 \cdot 11 \cdot 17$.
2. Let a be a formal variable, we have

$$(a-1)\sum_{i=0}^{n-1} a^i = \sum_{i=0}^{n-1} a^{i+1} - a^i = \sum_{i=1}^{n} a^i - \sum_{i=0}^{n-1} a^i = a^n - 1$$

3. By taking $a = a^m$ in the previous inequality we get the result.

4. Clearly $561 = 2 \cdot 280 + 1$ thus $a^{561} - a = a(a^{560} - 1) = a(a^{2 \cdot 280} - 1)$ and we have from the previous question by taking $m = 2$ and $n = 280$ that $a^{561} - a = a(a^2 - 1)\left(\sum_{i=1}^{279} a^{2i}\right) = a(a^2 - 1)m$.
5. Let $a \in \mathbb{Z}$, we have that $a = 3q + r$ with $0 \leq r < 2$ from the Euclidean division algorithm, then $a(a-1)(a+1) = (3q + r)(3q + (r-1))(3(q+1) + (r-2))$ and since $0 \leq r < 2$ then necessarily either r, $r - 1$ or $r - 2$ is 0, and thus it has one of the factors divisible by 3.
6. $a^{561} - a$ is divisible by 3 since $a^{561} - a = a(a+1)(a-1)m$ which is divisible by 3 from the previous question.
7. $a^{16} = 1 \bmod 17$ by Fermat little theorem, then $a^{560} = a^{16 \cdot 35} = 1 \bmod 17$ thus, $a^{561} = a \bmod 17 \Rightarrow 17 \mid a^{561} - a$.
8. Same here, $a^{10} = 1 \bmod 11$ by Fermat little theorem, then $a^{560} = a^{10 \cdot 56} = 1 \bmod 11$ thus, $a^{561} = a \bmod 11 \Rightarrow 11 \mid a^{561} - a$.
9. Given that all the divisors of 561 divide $a^{561} - a$ then it is divisible by their product so $561 \mid a^{561} - a \Rightarrow a^{561} \equiv a \bmod 561$.

Chapter 1

Introduction To Information Theory

The digital revolution has transformed how we communicate, store and process information. This chapter introduces the fundamental concepts of information theory, a discipline at the intersection of mathematics, computer science and telecommunication.

This diagram present the general structure of any communication system. The communication process begins with an information source. We start by thinking what a message could be. In general, we can consider a message to be constructed by a set of symbols $\mathcal{X} = \{x_1, ..., x_n\}$ and given a context, has a distribution of which of those symbols are sent. Thus, we consider a message consecutive realizations of a random variable X defined in $\mathcal{X}$.

Now the goal is to quantify information, thus given $X \sim P_X(x)$, we define the amount of information for some symbol x to be $I(x) = -\log_2(P_X(x))$ (justifications for these definitions will be in the next chapter). Then, we can define an average amount of information from a source X, which is called the entropy $H(X) = \mathbb{E}[I(X)]$. Such a quantity is important given Shannon's first theorem, which gives a lower bound for any data compression.

Notation: *Let Σ be a set:*
- $\Sigma^n = \Sigma \times \cdots \times \Sigma$ *is the set of words of length n. We take that $\Sigma^0 = \{\varepsilon\}$ where ε is the empty word.*
- $\Sigma^* = \cup_{i \geq 0} \Sigma^n$ *the set of all possible words over Σ.*
- $\Sigma^+ = \cup_{i \geq 1} \Sigma^n$ *the set of all possible non-empty words over Σ.*
- *For $x \in \Sigma^*$, we define $|x| = \min\{n \in \mathbb{N} \mid x \in \Sigma^n\}$.*
- *The set of binary letters as $\mathbb{B} = \{0, 1\}$.*
- *For $x = x_1...x_n, y = y_1...y_m \in \Sigma^*, xy = x_1...x_n y_1...y_m \in \Sigma^*$.*

Definition 1 (Symbol/Message): *Let $\mathcal{X} = \{x_1, ..., x_m\}$ be a finite set of alphabets. An element of $\mathcal{X}$ is called a symbol, and elements of $\mathcal{X}^+$ are called a message with alphabet $\mathcal{X}$.*

To model information that was given from a message, we consider the fact that a message contains more information the more unpredictable it is. For example, consider a sender that keeps sending the message "Hello!" over and over again, at each iteration, it becomes predictable what the next message would be, thus there is no more information that is passed. Uncertainty is a property of a random process, thus a good way to model a message is to consider it as a random variable X that has values in $\mathcal{X}$. From the previous example too, we can think that the information in a message is related to how uncertain it is, that is, how less probable it would be sent.

Notation: *For the sake of simplifying notation, we will denote $P(X = x)$ as $P(x)$, same for other random variables, like $P(Y = y)$ as $P(y)$. Same for all the remaining notations like $P(y|x)$ which is $P(Y = y|X = x)$.*

To define a measure for information $I : \mathcal{X} \to \mathbb{R}$, we start by considering some random variable X that represents a source of symbols $\mathcal{X}$. As discussed before, I would be inversely proportional to the probability $P(x)$ thus we have that

$$I(x) = f\left(\frac{1}{P(x)}\right)$$

f would satisfy the following properties:
- f is continuous, that is, a slight amount of information is gained from a slight difference in probability.
- $\lim_{t \to 1} f(t) = 0$, that is, the information given from a certain event is none.

- The amount of information given by a joint of two independent events x_1, x_2 is the sum of information from each of the events, thus $I(x_1, x_2) = I(x_1) + I(x_2)$.

$$I(x_1, x_2) = f\left(\frac{1}{P(x_1, x_2)}\right) = f\left(\frac{1}{P(x_1)} \cdot \frac{1}{P(x_2)}\right)$$
$$I(x_1, x_2) = I(x_1) + I(x_2) = f\left(\frac{1}{P(x_1)}\right) + f\left(\frac{1}{P(x_2)}\right)$$

Thus, f would satisfy $f(ab) = f(a) + f(b)$ which gives us that f is just a logarithm function. Therefore, we obtain the following definition below.

Definition 2 (Measure Of Information): *Let X be a random variable that represents the probability that a message x is sent as $P(X = x) = P(x)$, we define the measure of information of the message x as*

$$I(x) = -\log_2 P(x)$$

$\log_2$ is taken for the usual reason that information is represented in binary, and that if it is represented in any other base, it would have just a linear factor added. In practice, transmission lines tend to cause irregularities in the signal, that is, it alters the contents of the message with some noise. Thus, in transmission lines we are interested in the distribution of the message y that are received, given some sent message x, which will be measured by $P(y|x)$.

1.1. Entropy

After we defined a measure of information, which we discussed to be a measure of uncertainty, we need a way to quantify the average amount of uncertainty for all the values of X. Thus, we define the entropy as follows.

Definition 3 (Entropy): *Let X be a random variable with values in $\mathcal{X}$, we define the entropy $H(X)$ as the average amount of information, that is*

$$H(X) = \mathbb{E}[I(X)] = -\sum_{x \in X} P(x) \cdot \log_2 P(x)$$

Notice that the information of some message x depends only on its unpredictability, that is, its probability of occurrence $P(x)$ not its value.

Example: Let $\mathcal{X} = \{0, 1\}$, and X a discrete random variable with values in $\mathcal{X}$, which has the distribution

$$P(x) = \begin{cases} 0.9 \text{ if } x = 0 \\ 0.1 \text{ if } x = 1 \end{cases}$$

We obtain that

$$I(x) = \begin{cases} 0.152 \text{ if } x = 0 \\ 3.321 \text{ if } x = 1 \end{cases}$$

. And the measure of average amount of information is $H(X) = 0.4689$.

1.1.1. Source Coding

The most important part of coding theory is achieving the most efficient reliable and secure coding. The first part is the focus of this section, by assigning a code from each symbol in $\mathcal{X}$, we try to achieve the minimum bound possible of letters to send to transmit our message.

Definition 4 (Coding Function): *Let $c : \mathcal{X} \to \mathbb{B}^+$, we call it a coding function, which takes characters of our set of symbols, and represent it a binary string in $\mathbb{B}^+$.*

To measure the efficiency of our coding function, we define the average code length, the less the average, the more efficient the transmission will be.

Definition 5 (Average Code Length): *Let $c : \mathcal{X} \to \mathbb{B}^+$, a coding function, and consider the function*

$$\overline{L} : \{c : \mathcal{X} \to \mathbb{B}^+\} \to \mathbb{R}$$
$$c \mapsto \overline{L}_c = \sum_{x \in \mathcal{X}} p(x) \cdot |c(x)|$$

Example: Take a horse race with 8 horses, we want to send a message in binary that indicates which horse has won. And suppose that the probabilities of winning for each horse is as follows

Horse	1	2	3	4	5	6	7	8
Probability Of Winning	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$	$\frac{1}{64}$

If we just send the index of the winner horse, we get the following coding function

Horse i	1	2	3	4	5	6	7	8
$c_1(i)$	000	001	010	011	100	101	110	111

If we do it just blindly, we will need 3 bits to describe all the possible winner horses from $000, 001, 010, \cdots, 110, 111$, calculating the average of description length we get

$$\overline{L}_{c_1} = \sum_{x \in X} 3 \cdot p(x) = 3 \text{ bits}$$

Giving us an average of 3 bits for the transmission to give exactly who horse is the winner. An aspect that we did not use in the previous part is how likely do horses win in this case, which we can use to improve the amount of bits that would be sent on the channel. Consider the distribution of the horses winning as follows

Notice that if we give a smaller message for the horses that are more probable to win, and less to horses that are less likely to win, then we can reduce some of the data that will be used to specify the winner. We take the following encoding

Horse i	1	2	3	4	5	6	7	8
$c_2(i)$	0	10	110	1110	111100	111101	1111110	111111

Now, we recalculate the average description length to get

$$\overline{L}_{c_2} = \sum_{x \in X} |c(x)| \cdot p(x) = 2 \text{ bits}$$

we have reduced the average description length by 1 bit. If we calculate the entropy in this case we get $H(X) = -\sum_{x \in X} P(x) \cdot \log_2 P(x) = 1.83475$, we see that the reduced average code length is more than the entropy.

1.2. Shannon Theorems

To prove Shannon theorems, we need the following statements.

Definition 6 (Prefix Code): *A coding function $c : \mathcal{X} \to \mathbb{B}^+$ is said to have the prefix property if $\forall x, y \in \mathcal{X}, \forall u \in \mathbb{B}^*, c(x) \neq c(y)u$.*

That is, no code is a prefix of another. An example of a prefix code is $\{0, 10\}$ while a non-prefix code is $\{1, 10\}$ since 1 is a prefix of 10. These codes are also called instantaneous codes, that is because given a binary string, even incomplete, one can know if it maps to a symbol or not. The proof of such statement is left for the reader.

Definition 7 (Uniquely Decodable Code): *A coding function $c : \mathcal{X} \to \mathbb{B}^+$ is said to be uniquely decodable if c is injective, that is, each coding is unique to the symbol.*

Proposition 8: *A uniquely decodable code c can be turned into a prefix code c' such that $\forall i \in [\![1, n]\!], |c(x_i)| = |c'(x_i)|$.*

Lemma 9 (Kraft Inequality): *Let X be a random variable with values in $\mathcal{X}$, $c : \mathcal{X} \to \mathbb{B}^+$ a coding function and $l(x) = |c(x)|$, then we have that*

$$\sum_{i=1}^{n} 2^{-l(x_i)} \leq 1$$

Theorem 10 (Shannon's First Theorem): *Let X be a random variable with values in $\mathcal{X}$, for any coding function $c : \mathcal{X} \to \mathbb{B}^+$ we have*

$$\overline{L}_c \geq H(X)$$

Huffman's algorithm provides a systematic way to construct an optimal prefix code, where symbols with higher probability should have shorter code words, and achieves a lower bound of average coding weight.

Algorithm 10.1 (Huffman):

1. *Sort symbols by probability in descending order.*
2. *Combine the two smallest probabilities.*
3. *Repeat until one node remains.*
4. *Build the tree.*
5. *Assign 0 and 1 to branches.*
6. *Read codewords.*
7. *Compute the average code length.*

Example:

The closer this value is to the entropy, the more efficient the compression is, using Huffman coding, the message is compressed, meaning that fewer bits are needed on average per symbol compared to a fixed length code such code is called optimal code which is a code that minimizes average code length. In source coding, our goal was to remove redundancy in order to compress the message. Now, we move to a different problem, what happens when the channel is noisy?

1.2.1. Noise And Errors

To reduce the probability of error, we need to make the sent message more robust to disturbances through channel coding.

Theorem 11 (Shannon's Second Theorem): *For any noisy channel, there exists a code that allows reliable transmission if the rate is less than the channel capacity.*

Chapter 2

Information & Entropy

Information theory is a theory founded by Claude Shannon, providing a mathematical framework for quantifying information, studying its transmission, processing, and storage. This chapter focuses on the fundamental concepts of information measurement through entropy and related measures.

2.1. The Communication System Model

Shannon's groundbreaking contribution was a formal model that describes communication between two entities, whether machine on humans. This model highlights the essential components:

- Source: the originator of the information.
- Channel: The medium through which information is transmitted.
- Recipient: The receiver of the information.

2.1.1. Source & Message

The sources generate a message composed of symbols from a specific alphabet.

- Discrete data: text, numbers, symbols.
- Digitized analog data: image, voice, video converted to digital form.

2.1.2. Transmitter Processing

The transmitter performs three crucial operations.

1. Source Coding: data compression to minimize message size.
2. Channel Coding: adding redundancy for error correction.
3. Modulation: converting digital messages to physical signal.

2.1.3. Receiver Processing

The receiver performs these receiver operations.

1. Demodulation: converting physical signal into a digital message.
2. Channel Decoding: error detection and correction.
3. Source Decoding: reconstruction of the original message.

2.2. Quantifying Information

Shannon's revolutionary idea was to assign a measurable value to information based on probability rather than meaning or semantics.

2.2.1. Fundamental Postulates Of Information

- Postulate 1: Information depends only on probability $I(A) = \Psi(P(A))$.
- Postulate 2: Certain events provide no information $\Psi(1) = 0$.
- Postulate 3: Information decreases with increasing probability.
- Postulate 4: Impossible events provide infinite information $P(0) = \infty$.
- Postulate 5: Information from independent events $\Psi(pq) = \Psi(p) + \Psi(q)$.

Which gives the function $\Psi(x) = -\log_2 x$, thus, the self-information of an event A with probability $P(A)$ is $I(A) = -\log_2(P(A))$.

2.3. Entropy

While self-information measures the information content of individual events, we often need to characterize the average information by a source.

2.3.1. Definition Of Entropy

For a discrete random variable X with probability mass function $p(x)$, the entropy is defined as

$$H(X) = E[I(x)] = -\sum_{x \in X} p(x) \log_2(p(x))$$

Example: Consider a binary source with $P(0) = p$ and $P(1) = 1 - p$ then $H(X) = -p \log_2(p) - (1-p) \log_2(1-p)$.

2.3.2. Properties Of Entropy

1. Non-negativity: $H(X) \geq 0$.
2. Maximum entropy: for a source with symbols in $\mathcal{X}$, $H(X) \leq \log_2 \# \mathcal{X}$.
3. Positivity for independent sources: if X, Y are independent then $H(X, Y) = H(X) + H(Y)$.

Example: DNA sequences provide an excellent example of entropy application. Let's consider:

- Primate DNA: $P(A) = P(G) = P(C) = P(T) = 1/4$. $H(X) = 2$
- Bacterial DNA: $P(G) = P(C) = 0.38$, $P(A) = P(T) = 0.12$. $H(X) = 1.795$.

The entropy of bacterial DNA indicates more predictable sequences compared to the primate DNA.

2.3.3. Joint Entropy

When dealing with random variables, we need to quantity combined information.

> **Definition (Joint Entropy):** *Let X, Y be two random variables with joint distribution $p(x, y)$, the joint entropy is*
>
> $$H(X, Y) = -\sum_{x \in X} \sum_{y \in Y} p(x, y) \log_2 p(x, y)$$

2.3.4. Properties Of Joint Entropy

1. Symmetry: $H(X, Y) = H(Y, X)$.
2. Non-Negativity: $H(X, Y) \geq 0$.
3. Upperbound: $H(X, Y) \leq H(X) + H(Y)$.
4. Lowerbound: $H(X, Y) \geq \max(H(X), H(Y))$.